Exp No: 9

Arduino based manual electronic counter using proteus

AIM:

To write an Embedded C program for Manual Electronic counter using Arduino IDE and Proteus

SOFTWARE REQUIRED:

• Proteus 8 software

• Arduino IDE

PROGRAM:

```
int x0,x1,x2,x3,x4,x5,x6,x7,x8,x9;
int delay_time=200;

void setup() {
//configure pin2 as an input and enable the internal pull-up resistor
pinMode(12, INPUT_PULLUP);
pinMode(1,OUTPUT);
pinMode(2,OUTPUT);
pinMode(3,OUTPUT);
pinMode(4,OUTPUT);
pinMode(5,OUTPUT);
pinMode(6,OUTPUT);
pinMode(7,OUTPUT);

}
void loop() {
//read the pushbutton value into a variable
int sensorVal = digitalRead(12);

if (sensorVal == LOW) {
```

```
x0=true;
}
while(x0){
zero();
sensorVal = digitalRead(12); if
(sensorVal == LOW) {
x1=true;
x0=false;
}
}
while(x1){
one();
sensorVal = digitalRead(12); if
(sensorVal == LOW) {
x2=true;
x1=false;
}
}
while(x2){
two();
sensorVal = digitalRead(12); if
(sensorVal == LOW) { x3=true;
x2=false;
}
}
while(x3){
three();
sensorVal = digitalRead(12); if
(sensorVal == LOW) { x4=true;
```

```
x3=false;


}
}
while(x4){
four();
sensorVal = digitalRead(12); if
(sensorVal == LOW) { x5=true;
x4=false;
}
}
while(x5){
five();
sensorVal = digitalRead(12); if
(sensorVal == LOW) { x6=true;
x5=false;
}
}
while(x6){
six();
sensorVal = digitalRead(12); if
(sensorVal == LOW) { x7=true;
x6=false;
}
}
while(x7){
seven();
sensorVal = digitalRead(12); if
(sensorVal == LOW) { x8=true;
```

```
x7=false;
}
}
while(x8){

eight();
sensorVal = digitalRead(12); if
(sensorVal == LOW) { x9=true;
x8=false;
}
}
while(x9){
nine();
sensorVal = digitalRead(12);
if (sensorVal == LOW) {
x0=true;
x9=false;
}
}

}

void zero()
{
for(int i=1;i&lt;7;i++)
{
digitalWrite(i,HIGH);
digitalWrite(7,LOW);
}
```

```
delay(delay_time);
}

void one()
{
digitalWrite(1,LOW);
digitalWrite(2,HIGH);

digitalWrite(3,HIGH);
digitalWrite(4,LOW);
digitalWrite(5,LOW);
digitalWrite(6,LOW);
digitalWrite(7,LOW);
delay(delay_time);
}
void two()
{
digitalWrite(1,HIGH);
digitalWrite(2,HIGH);
digitalWrite(3,LOW);
digitalWrite(4,HIGH);
digitalWrite(5,HIGH);
digitalWrite(6,LOW);
digitalWrite(7,HIGH);
delay(delay_time); }

void three()
{
digitalWrite(1,HIGH);
```

```
digitalWrite(2,HIGH);
digitalWrite(3,HIGH);
digitalWrite(4,HIGH);
digitalWrite(5,LOW);
digitalWrite(6,LOW);
digitalWrite(7,HIGH);
delay(delay_time); }
void four()
{
digitalWrite(1,LOW);
digitalWrite(2,HIGH);
digitalWrite(3,HIGH);
digitalWrite(4,LOW);
digitalWrite(5,LOW);
digitalWrite(6,HIGH);
digitalWrite(7,HIGH);
delay(delay_time);
}
void five()
{
digitalWrite(1,HIGH);
digitalWrite(2,LOW);
digitalWrite(3,HIGH);
digitalWrite(4,HIGH);
digitalWrite(5,LOW);
digitalWrite(6,HIGH);
digitalWrite(7,HIGH);
delay(delay_time); }
void six()
```

```
{
digitalWrite(1,HIGH);
digitalWrite(2,LOW);
digitalWrite(3,HIGH);
digitalWrite(4,HIGH);
digitalWrite(5,HIGH);
digitalWrite(6,HIGH);
digitalWrite(7,HIGH);
delay(delay_time);
}
void seven()
{
digitalWrite(1,HIGH);
digitalWrite(2,HIGH);
digitalWrite(3,HIGH);
digitalWrite(4,LOW);
digitalWrite(5,LOW);
digitalWrite(6,LOW);
digitalWrite(7,LOW);
delay(delay_time); }
void eight()
{
digitalWrite(1,HIGH);
digitalWrite(2,HIGH);
digitalWrite(3,HIGH);
digitalWrite(4,HIGH);
digitalWrite(5,HIGH);
digitalWrite(6,HIGH);
digitalWrite(7,HIGH);
```

```
delay(delay_time);
}
void nine()
{
digitalWrite(1,HIGH);
digitalWrite(2,HIGH);
digitalWrite(3,HIGH);
digitalWrite(4,HIGH);
digitalWrite(5,LOW);
digitalWrite(6,HIGH);
digitalWrite(7,HIGH);
delay(delay_time);
}
```

OUTPUT: